

RL Assignment - 4

2303A510H1
(Batch 14)

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Upper Confidence Bound for Multi-Armed Bandit

Q) Consider three slot machines A, B and C. Each machine provides a numerical reward whenever it is played. The objective is to use the Upper Confidence Bound algorithm to select a machine over 10 plays.

* UCB Formula

$$UCB(a) = Q(a) + \sqrt{\frac{2 \ln(t)}{N(a)}}$$

Where: $Q(a)$ is the estimated value of machine a .

t is the current play number

$N(a)$ is the number of times machine a has been selected.

Incremental Sample Average Formula

$$Q_{\text{new}}(a) = Q_{\text{old}}(a) + \frac{1}{N(a)} (R - Q_{\text{old}}(a))$$

$Q_{\text{old}}(a)$ is the current estimated value.

R is the reward obtained.

$N(a)$ is the number of times the machine has been selected.

Calculations for Two Plays

* Play 1

$$N(A) = N(B) = N(C) = 0$$

$$Q(A) = Q(B) = Q(C) = 0$$

$$R = 3$$

$$N(A) = 1$$

$$Q(A) = 0 + \frac{1}{1} (3 - 0) = 3$$

∴ after Play 1

$$N(A) = 1$$

$$Q(A) = 3$$

* Play 2

$$R = 6$$

$$N(B) = 1, Q(B) = 0$$

$$Q(B) = 0 + \frac{1}{2} (6 - 0)$$

$$Q(B) = 3$$

∴ after Play 2 :

$$N(B) = 1$$

$$Q(B) = 3$$